

Module 4 — Variant Calling: Runbook

This is an **operational** document — step-by-step execution instructions, expected outputs, and troubleshooting. For the conceptual "why," see this module's [README.md](#). For the full curriculum context, see [docs/curriculum/modular_bioinformatics_curriculum.md](#).

Prerequisites

- Slurm environment with module load access to SRA-Toolkit/3.2.0-gompi-2024a, fastp/0.23.4-GCC-13.2.0, BWA/0.7.18-GCC-13.3.0, SAMtools/1.21-GCC-13.3.0, BCFtools/1.21-GCC-13.3.0, and MultiQC/1.28-foss-2024a — used by Tier A (master_snp_pipeline.sh).
- samtools and bcftools on PATH (outside Slurm too) for Tier B Scripts 1, 2, 3, 4, and 6 — these are plain shell scripts, not Slurm jobs, and are meant to be run interactively against Tier A's output.
- R with vcfR, adegenet, and ape installed — used by Tier A's variant_analysis.R and Tier B Scripts 5 and 7.
- Module 1 completed or reviewed — this module assumes the same "validate/interpret before you trust it" habit taught there, applied one level up the pipeline (aligned reads and called variants instead of raw reads).

A real, verified data-provenance issue (read before running the smoke test)

master_snp_pipeline.sh's own built-in fallback srr_list.txt (used only when no srr_list.txt is staged) is SRR11092056/SRR11092057. Verified via NCBI eutils: SRR11092056 is SARS-CoV-2 metagenomic RNA-seq (Wuhan Institute of Virology), not bacterial WGS, and cannot produce a meaningful alignment against this pipeline's Typhimurium LT2 reference. **Do not rely on the built-in fallback.** Use smoke_test_prjna242847.sh, which stages a real, verified 3-isolate Typhimurium cohort (PRJNA242847) before invoking master_snp_pipeline.sh. See the README's data-provenance note for the full verification trail.

Execution order

Step	Command	Expected result
1	bash smoke_test_prjna242847.sh (or sbatch master_snp_pipeline.sh with your own verified srr_list.txt)	variants/<SRR>_raw.vcf, variants/<SRR>_filtered.vcf, variants/merged_cohort.vcf
2	bash 1_Validate_Alignment_And_VCF_Inputs.sh [WORKDIR]	Console PASS/FAIL/WARN report per sample; stops hard on a structural or name-mismatch failure
3	bash 2_Assess_Mapping_And_Reference_Bias.sh [WORKDIR]	mapping_bias_summary.tsv; console table of % mapped, % properly paired, mean depth, % zero-coverage per sample
4	bash 3_Design_Defensible_Variant_Filter.sh [WORKDIR]	variants/<SRR>_filtered_v2.vcf, variants/<SRR>_strand_biased_sites.tsv; console naive-vs-v2 PASS-count comparison
5	bash 4_Check_Ploidy_Assumptions.sh [WORKDIR]	variants/<SRR>_raw_haploid.vcf; console before/after heterozygous-call count per sample
6	Re-merge the ploidy-corrected calls: cd WORKDIR/variants && for f in *_raw_haploid.vcf; do bcftools view -Oz -o \${f%.vcf}.vcf.gz \$f && bcftools index -t \${f%.vcf}.vcf.gz; done && bcftools merge *_raw_haploid.vcf.gz -Ov -o merged_cohort_haploid.vcf	variants/merged_cohort_haploid.vcf — the ploidy-correct cohort VCF

Step	Command	Expected result
7	<code>Rscript 5_Cohort_VCF_QC_Summary.R WORKDIR/variants/merged_cohort_haploid.vcf</code>	<code>cohort_vcf_qc_summary.tsv</code> , <code>cohort_vcf_qc_per_sample.tsv</code> ; console Ti/Tv and heterozygosity report
8	<code>bash 6_Coverage_Gaps_And_Limitations.sh [WORKDIR]</code>	<code>LIMITATIONS.md</code>
9	<code>cd WORKDIR/variants && Rscript variant_analysis.R</code> (Tier A, against <code>merged_cohort_haploid.vcf</code> — see script header for the required <code>bcftools merge</code> setup if not reusing step 6's output)	<code>variant_quality_qc.pdf</code> , <code>phylogenetic_tree.pdf</code> , <code>outbreak_clusters.tsv</code> (if any pair is within <code>SNP_THRESHOLD</code>)
10	<code>Rscript 7_Assemble_Variant_Report.R WORKDIR</code>	<code>VARIANT_REPORT.md</code> , sections 1-7 auto-filled, section 8 blank
11	Fill in Section 8 (interpretation) of <code>VARIANT_REPORT.md</code> by hand	Completed portfolio artifact

Step 6 is the one manual hand-off in this module: Script 4 corrects the *calling* step but does not silently rewrite Tier A's cohort merge for you — running Scripts 5-7 against the original `merged_cohort.vcf` instead of the ploidy-corrected re-merge will produce numbers Script 5 itself will flag (non-zero heterozygosity against a haploid reference), not numbers that are silently wrong.

Expected outputs, by step

- **Step 1 (Tier A):** `alignment/<SRR>_dedup.bam[.bai]`, `variants/<SRR>_raw.vcf`, `variants/<SRR>_filtered.vcf`, `variants/merged_cohort.vcf`, `final_multiqc_report.html`.
- **Step 2:** stdout only — validation report.
- **Step 3:** `mapping_bias_summary.tsv`.
- **Step 4:** `variants/<SRR>_filtered_v2.vcf`, `variants/<SRR>_strand_biased_sites.tsv`, `variants/<SRR>_raw_metrics.tsv`.
- **Step 5:** `variants/<SRR>_raw_haploid.vcf`.
- **Step 6:** `variants/merged_cohort_haploid.vcf`.
- **Step 7:** `cohort_vcf_qc_summary.tsv`, `cohort_vcf_qc_per_sample.tsv`.
- **Step 8:** `LIMITATIONS.md`.
- **Step 9 (Tier A):** `variant_quality_qc.pdf`, `phylogenetic_tree.pdf`, `outbreak_clusters.tsv` (conditional).
- **Step 10:** `VARIANT_REPORT.md`.

Troubleshooting

Symptom	Likely cause	Fix
<code>smoke_test_prjna242847.sh</code> fails at the <code>prefetch/fasterq-dump</code> step	SRA-Toolkit module not loaded, or a transient NCBI connectivity issue	Confirm module load <code>SRA-Toolkit/3.2.0-gompi-2024a</code> succeeded; retry — <code>master_snps_pipeline.sh</code> 's per-sample loop is fault-tolerant and will skip a genuinely failed sample rather than abort the whole run
Script 1 fails with <code>@RG SM: tag</code> does not match filename-derived sample	A BAM was renamed/copied after alignment without regenerating its read group, or two samples' outputs were cross-copied	Re-run alignment for that sample from <code>master_snps_pipeline.sh</code> rather than renaming files by hand — the <code>-R "@RG\tID:...\tSM:..."</code> flag is what makes this checkable at all
Script 2 flags most/all samples <code>LOW_MAPPING_RATE</code>	You are accidentally running this against the pipeline's own broken default accessions (SARS-CoV-2 RNA-seq vs. a bacterial DNA reference) instead of a verified worked-example cohort	Confirm <code>srr_list.txt</code> matches <code>smoke_test_prjna242847.sh</code> 's accessions or your own verified bacterial WGS list — see the README's data-provenance note

Symptom	Likely cause	Fix
Script 3 reports 0 strand-biased sites for every sample	Either genuinely clean data, or too few candidate calls with INFO/DP4 ALT depth ≥ 4 to evaluate (very shallow coverage)	Cross-check against Script 2's mean-depth column before treating "0 flagged" as "no strand bias present"
Script 4 reports 0 heterozygous calls under the diploid default	Does not confirm ploidy was handled correctly — it means this particular run got lucky, not that the risk doesn't exist. The *_raw_haploid.vcf files are still the correct calls to build on	Always continue with the haploid-corrected files regardless of whether this run happened to show 0 het calls
Script 5 reports non-zero heterozygosity	You ran it against Tier A's original merged_cohort.vcf (diploid-default calls) instead of the ploidy-corrected re-merge from Step 6	Re-merge *_raw_haploid.vcf per Step 6 and re-run Script 5 against that file
Script 5 reports Ti/Tv ≤ 0.6	Cohort's SNP calls skew toward random noise rather than real biological signal — could be genuinely low-quality data, or evidence you are still running against the wrong (viral RNA-seq vs. bacterial reference) dataset	Confirm the dataset first (see the mapping-rate check above); if the dataset is correct, treat this as a real quality warning, not something to filter past silently
variant_analysis.R (Tier A) errors on read.vcfR("merged_cohort.vcf")	Working directory is not WORKDIR/variants, or the file wasn't regenerated after Step 6's re-merge	cd into WORKDIR/variants first; confirm the intended cohort VCF (original vs. haploid-corrected) actually exists there under the name the script expects
Script 7's VARIANT_REPORT.md shows "Not available" for a section	The corresponding earlier script was not run, or was run in a different working directory than the one passed to Script 7	Re-run the missing step with the same WORKDIR argument, then re-run Script 7
Script 7's Section 8 is blank	Expected — intentionally left for the analyst to complete by hand	Fill it in based on the actual mapping/coverage flags (Section 2), Ti/Tv (Section 5), and clustering results (Section 6) — not the auto-generated prompt text

Completion checklist (portfolio artifact)

The curriculum's Module 4 assessment asks for: *"A filtered VCF + a QC summary + an explicit statement of the call set's limitations (coverage gaps, reference bias, filter thresholds used)."* Confirm each is present before considering the module complete:

- [] **Filtered VCF** — <SRR>_filtered_v2.vcf (Script 3) or, better, the ploidy-corrected merged_cohort_haploid.vcf (Step 6) is retained alongside the report, not just summarized.
- [] **QC summary** — Section 5 of VARIANT_REPORT.md reflects a real Script 5 run against your actual merged cohort VCF, including the Ti/Tv value and per-sample heterozygosity, not a placeholder.
- [] **Coverage gaps / reference bias** — Section 2 reflects a real Script 2 run (per-sample mapping rate and % zero-coverage), not a placeholder.
- [] **Filter thresholds used** — Section 3 states the exact per-sample thresholds Script 3 actually computed for this cohort, not the naive fixed defaults alone.
- [] **Ploidy status** — Section 4 states plainly whether the reported calls are diploid-default or ploidy-corrected (Script 4), and Section 5's heterozygosity numbers are consistent with that answer.
- [] **Interpretation** — Section 8 is filled in by hand, referencing the actual mapping/coverage, filter, ploidy, and Ti/Tv findings above — not left as the auto-generated prompt text.